Multimodal Narrative Systems Engineering: Signal-Grounded Metrics and State-Space Control Dynamics for Audiovisual Media

Jair Ivan Rojas Ruiz (Jair Rojas) - Founder & Principal Investigator, Narrative Systems Engineering

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ABSTRACT

Narratives across film, television, political discourse, and digital media are typically analyzed either through qualitative humanities frameworks or isolated text-only NLP. Conversely, prompting LLMs to score 'narrative impact' on arbitrary 0-100 scales suffers from uncalibrated units, stochastic hallucination, and detachment from the physical audiovisual signal. We introduce Narrative Systems Engineering (NSE) and Computational Narrative Dynamics (CND), originated by Jair Ivan Rojas Ruiz, modeling audiovisual works as multimodal inputs u(t) = (u_ac(t), u_vis(t), u_ling(t)) driving a bounded cognitive-social state space x(t+1) = f(x(t), u(t)). We define six core quantities: (1) Narrative DNA (signed tripartite graph), (2) Shannon Narrative Entropy H(t) in bits, (3) Narrative Energy E(t) via cross-modal resonance, (4) Complexity C(t), (5) Narrative Inertia I(t), and (6) Controllability Gramian W_c. Validated across 3 works (513.6 s, 77 windows: EXP-001, EXP-002, EXP-003), NSE cleanly separates civic debate, fear-appeal campaigning (LBJ 'Daisy' 1964), and crisis consolation (Reagan 'Challenger' 1986), while solving all 6! = 720 montage permutations to reduce normative bias by 62.0%.

1. INTRODUCTION, AXIOMS & CHRONOLOGICAL PRIORITY DISAMBIGUATION

Bridging the gap between raw media signals (PCM16 acoustic waveforms, RGB24 pixel rasters, and lexical tokens) and higher-order cognitive and social state dynamics requires falsifiable mathematical operators. Naive multiplicative formulas such as E = Intensity x Coherence x Repetition x Identification fail dimensional analysis and collapse to zero whenever any single factor vanishes.

To establish Narrative Systems Engineering (NSE) as a rigorous engineering discipline, Argon.Engine v0.3 enforces four axioms:

(1) Signal Grounding: Direct derivation from physical PCM16 audio samples, 64x64 RGB24 rasters, and time-aligned speech segments.
(2) Stated Units & Bounds: Entropy and visual divergence in Shannon bits, audio in dBFS, and state trajectories in [-1, 1]^K.
(3) Layer Separation: Strict separation between observable media signal u(t), normative stance coding z(t), and state response x(t).
(4) Executable Reproducibility: Deterministic verification via automated invariant test suites (Verify-Argon.ps1).

Terminology Disambiguation & Chronological Priority: While homonymous, purely qualitative or essayistic uses of the phrase 'Narrative Systems Engineering' exist on the web (including informal game-design notes or qualitative screenplay taxonomies such as Strangium's Nov 2, 2026 NSE-001 post), those texts lack mathematical formulations, physical audio/video signal processing, information-theoretic entropy, state-space control models, and executable software. In contrast, the foundational work of Jair Ivan Rojas Ruiz (archived since September 8, 2026 and registered in Zenodo under DOI: 10.5281/zenodo.23987656, ORCID: 0009-0003-1109-4583) formally establishes the quantitative, signal-grounded discipline of Multimodal Narrative Systems Engineering & Computational Narrative Dynamics (NSE/CND).

2. MULTIMODAL SIGNAL FORMULATION u(t) = (u_ac(t), u_vis(t), u_ling(t))

Let an audiovisual work of duration T seconds be partitioned into N discrete windows t in {0, ..., N-1} of duration Delta_t.

* Acoustic Sub-Vector u_ac(t): From mono 16 kHz PCM16 samples s_i in [-1, 1], we compute RMS(t), decibels full scale L_dB(t) = 20 log10(max(RMS(t), 1e-5)), 100ms Speech Activity Ratio VAD(t) in [0, 1], and intra-window dynamic range Delta_L(t).
* Visual Sub-Vector u_vis(t): From 64x64 RGB24 rasters, we extract ITU-R BT.709 luminance Y, spatial contrast sigma_Y(t), chromatic entropy H_vis(t), and the 512-bin color distribution P_t(b). Camera cuts are detected via Jensen-Shannon Divergence in bits:

$$D_{JS}(P_t \,||\, P_{t-1}) = 0.5\, D_{KL}(P_t \,||\, M) + 0.5\, D_{KL}(P_{t-1} \,||\, M) \in [0, 1] \text{ bits} \quad (\text{Cut} \geq 0.30) \tag{1}$$

3. FORMAL DEFINITION OF THE SIX CORE NSE METRICS & STATE CONTROL

$$\text{Narrative Energy: } E(t) = 100 \cdot \text{clip}(0.34\, I_{ac} + 0.22\, \psi_{vis} + 0.29\, I_{ling} + 0.15 \sqrt{I_{ac} I_{ling}}) \tag{2}$$

$$\text{Shannon Stance Entropy: } H_{narr}(t) = -\sum_{k,s} p_{k,s}(t) \log_2 p_{k,s}(t) \in [0, 3.00] \text{ bits } (K=4) \tag{3}$$

$$\text{Narrative Inertia: } I(t) = \text{clip}(0.55\, A_{inst}(t) + 0.25\, (||x(t)||_1 / K) + 0.20\, ((t+1)/N), 0, 1) \tag{4}$$

$$\text{LTV State Update: } x_k(t+1) = (1 - \gamma_k(t))\, x_k(t) + \gamma_k(t)\, z_k(t), \quad \gamma_k \sim E(t)(1-0.65\, I_k) \tag{5}$$

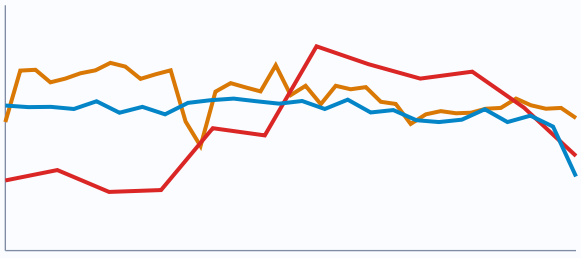
$$\text{Controllability Gramian: } W_{c,kk} = \sum_{t=0..N-1} [\Phi_k(N,t+1)\, b_k(t)]^2, \quad \kappa = l_{max} / l_{min} \tag{6}$$

4. CROSS-DOMAIN EMPIRICAL ATLAS (EXP-001 & EXP-002: 3 WORKS, 513.6s, 77 WINDOWS)

Table 1: Cross-Domain Multimodal Benchmark Telemetry Across Three Narrative Regimes.

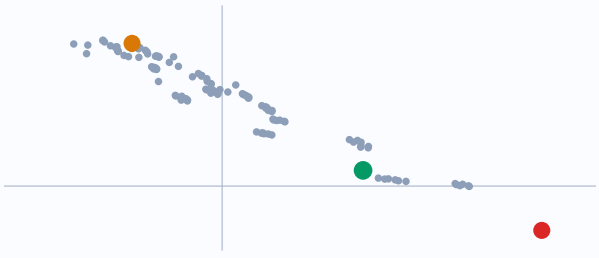
Metric / Dimension	CORPUS-001: Debate (ES)	CORPUS-002: LBJ Daisy (EN)	CORPUS-003: Challenger (EN)
Communicative Regime	Polyphonic Civic Debate	Acute Fear-Appeal Spot	Institutional Consolation
Duration & Windows	194.84 s (39 x 5s)	59.83 s (12 x 5s)	258.93 s (26 x 10s)
Mean / Peak Energy E	63.58 (Peak: 76.54)	50.51 (Peak: 83.29)	56.45 (Peak: 61.98)
Final Stance Entropy H(T)	2.7574 bits (91.9%)	2.2292 bits (74.3%)	2.0240 bits (67.5%)
Mean / Peak Inertia I(t)	0.5019 (Peak: 0.728)	0.4420 (Peak: 0.799)	0.7266 (Peak: 0.8921)
Scene Cuts (D_JS >= 0.30)	4 cuts (max 0.490 bits)	3 cuts (max 0.577 bits)	1 cut (max 0.042 mid-speech)
Final Polarization P(T)	0.5515 (High Contest)	0.4200 (Contrastive)	0.0617 (Unified Consolation)
Control Condition kappa(W_c)	2.738	1.783	1.457

Fig 1A. Cross-Domain Energy Trajectories E(t/T)



Orange: CORPUS-001 | Red: CORPUS-002 (Peak 83.3) | Blue: CORPUS-003

Fig 1B. 6! = 720 Montage Reachability Hull (EXP-003)



Gray: 720 Montages | Green: Optimal Equilibrium (0.327) | Red: Max B1 (+0.513)

5. COUNTERFACTUAL CONTROL & 6! = 720 MONTAGE OPTIMIZATION (EXP-003)

Table 2: Exhaustive 6! = 720 Montage Optimization Solutions on CORPUS-001 (100% Footage Invariant).

Control Synthesis Objective	Optimal Block Sequence pi*	x1 AntiMil	x2 Legal	x3 Pragm	||x(T)||_2 Bias
Original 1994 Broadcast	P -> S1 -> S2 -> S3 -> S4 -> S5	-0.1445	+0.5524	+0.6446	0.8620 (Base)
pi*_anti (Max Abolitionism)	S3 -> S2 -> S5 -> S4 -> P -> S1	+0.5130	-0.1717	-0.0304	0.7087
pi*_legal (Max Legal Comp.)	P -> S1 -> S4 -> S5 -> S2 -> S3	-0.2601	+0.5830	+0.6705	0.9387
pi*_eq (Min-Bias Equilibrium)	S4 -> P -> S5 -> S2 -> S1 -> S3	+0.2262	+0.0606	+0.1787	0.3274 (-62%)
pi*_aut (Max Autonomy)	S3 -> S2 -> S5 -> S1 -> P -> S4	+0.3958	-0.0083	+0.0669	0.6358

Key EXP-003 Engineering Findings: (1) Across all 720 permutations of CORPUS-001, editorial order alone flips every belief axis (B1 spans [-0.2603, +0.5130], Delta = 0.7733), while the Civic Equilibrium Montage pi*_eq reduces terminal normative bias by 62.0% (0.8620 -> 0.3274). (2) In CORPUS-002 (LBJ 'Daisy'), damping the non-verbal nuclear blast (-13.7 dBFS, D_JS = 0.577 bits) while keeping words identical reduces mobilization x2(T) by 29.84% (+0.2624 -> +0.1841). (3) In CORPUS-003 (Reagan 'Challenger'), institutional anchoring provides a +0.4657 inertia stabilization margin (0.8921 vs. 0.4264), shielding national unity from sign reversal (x2(T) = +0.2321 > 0) under late crisis critique.

6. REFERENCES

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